

Worksheet 4: Section 1.4

Quadratic Equations

Name: _____

Date: _____

Key Concepts

Quadratic Equations:

- **Zero Product Property:** If $mn = 0$, then $m = 0$ or $n = 0$.
- **Square Root Property:** If $x^2 = k$, then $x = \pm\sqrt{k}$.
- **Completing the Square:** To solve $ax^2 + bx + c = 0$:
 1. Divide both sides by a to make the leading coefficient 1.
 2. Isolate the variable terms on one side: $x^2 + bx = -c$
 3. Complete the square: Add $\left(\frac{b}{2a}\right)^2$ to both sides.
 4. Factor the left side as a perfect square: $\left(x + \frac{b}{2}\right)^2 = \text{constant}$
 5. Apply the square root property and solve for x .
- **Quadratic Formula:** $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Discriminant:** $D = b^2 - 4ac$ determines the number and type of solutions:
 - $D > 0$: 2 real solutions
 - $D = 0$: 1 real solution
 - $D < 0$: 2 complex solutions

Practice Problems

Show all work. Box your final answers.

Problem 1. Solve using the Zero Product Property:

$$x(x - 1) = 0$$

Solution:

Problem 2. Solve by factoring:

$$x(2x + 1) = 15$$

Solution:

Problem 3. Solve by factoring:

$$24x^2 = 8x$$

Solution:

Problem 4. Solve using the Square Root Property:

$$3x^2 = 12$$

Solution:

Problem 5. Solve using the Square Root Property:

$$(x - 5)^2 = 7$$

Solution:

Problem 6. Solve by completing the square:

$$x^2 - 18x + 5 = 0$$

Solution:

Problem 7. Solve using the Quadratic Formula:

$$2x(x - 3) = -7$$

Solution:

Problem 8. Determine the number and type of solutions for:

$$5x^2 - 3x + 1 = 0$$

Solution:

Answer Key

Answer Key

Problem 1: $x = 0$ or $x = 1$

Problem 2: $x = \frac{5}{2}$ or $x = -3$

Problem 3: $x = 0$ or $x = \frac{1}{3}$

Problem 4: $x = \pm 2$

Problem 5: $x = 5 \pm \sqrt{7}$

Problem 6: $x = 9 \pm 2\sqrt{19}$

Problem 7: $x = \frac{3 \pm i\sqrt{5}}{2}$

Problem 8: $D = -11 < 0$, so 2 complex solutions